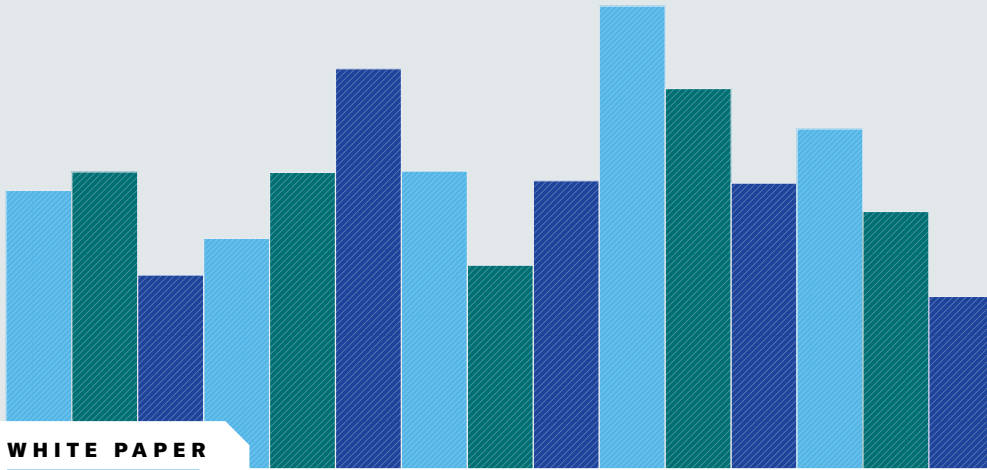




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WHITE PAPER

Embracing Modern Software Development Practices in the AI Era



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In today's fast-paced business environment, the ability of enterprises to rapidly develop software applications is not just a competitive advantage but a necessity. The pressure to deliver more software at a faster pace is at an all-time high, with significant financial and reputational risks at stake for companies that fail to keep up.

Technology innovation is supporting the demand for rapid software development. The promise of low-code/no-code platforms is being realized, allowing a new group of users across a business to create any tech solution they need—no coding skills required. Projects that once took months now happen in days, fueled by solutions and apps that can be built in minutes.

Such speed and efficiency in software development are crucial for businesses to innovate, adapt, and thrive in the modern digital landscape. However, this ease and speed introduces new challenges. What was once a path to greater productivity has now opened the door to a new kind of operational chaos. Businesses must be capable and confident in their ability to quickly develop and deploy high-quality custom artificial intelligence (AI)-powered solutions with advanced governance controls to ensure the business isn't slowed by security concerns.

In the race to innovate faster than ever before, businesses face critical questions: Can speed and control coexist? Can effective and comprehensive monitoring take place across hybrid multicloud infrastructures? Can quality be ensured given the proliferation of application programming interfaces, snowballing generative AI tools, and downward budget pressure on team sizes? Can governance be deployed at scale to ensure security standards and compliance regulations are met inclusive of the software supply chain?

The answer to each of these questions is yes for those that employ an approach detailed in this white paper researched by Harvard Business Review Analytic Services. Informed by software development leaders and experts from Fortune 500 companies, global consulting firms, and technology research institutions, this report highlights four interrelated pillars of modern software development.

When properly deployed, this approach can consistently result in the rapid innovation and delivery of high-quality, optimized, secure, and compliant software. Even more encouraging is that AI-driven solutions are available to support, manage, and ensure the success of each pillar.

Together with AWS, LogicMonitor, Quickbase, SmartBear, and Sonatype have come together to sponsor this research and white paper to equip and empower teams to navigate the changing landscape with confidence. We encourage you to read on to understand the four interrelated pillars and then consider how you can deploy and support them in your business.



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Embracing Modern Software Development Practices in the AI Era

In an era of artificial intelligence (AI) and rapidly changing business requirements, enterprises must accelerate their software delivery even while ensuring their development processes accommodate the inevitable need for more code, infrastructure, and testing.

The evolving digital landscape demands modern software development practices that encompass four interrelated pillars: speed and agility, improved visibility through testing and observability, AI-powered automation, and embedded security and governance.

Done correctly, this approach can empower companies to innovate, adapt, and thrive in a time when speed is paramount. By 2028, software development will have undergone a radical transformation, enabling teams to build new applications at previously unimaginable speeds, according to Forrester Research Inc., headquartered in Cambridge, Mass.

Diego Lo Giudice, vice president and principal analyst at Forrester, explains, “Software is the expression of a business, since it provides the foundation for all of the organization’s processes, policies, and services. Creating a continuous, iterative software delivery model depends on having an infrastructure that is designed from soup to nuts to speed up the development process.”

Incorporating generative AI (gen AI) into these processes can have a profound effect on IT personnel. A study by McKinsey & Co. found that 87% of developers who used gen AI said they were able to focus on meaningful and satisfying work, compared to only 50% of developers who did not use the technology. **FIGURE 1**

And this effect on IT personnel underscores, like any profound technology change, that modern software development is about far more than technology. The four pillars require heightened levels of collaboration. Highly skilled workers must be empowered to innovate, even as guardrails are put in place to protect corporate assets. Executives must rethink practices such as how they budget IT in order to ensure software development is not bogged down

HIGHLIGHTS

Modern software development is undergoing profound change as organizations **adopt new practices to provide the speed, agility, and innovation** that is required by the changing business world.

The transformation requires a shift in **organizational commitment to continuous learning and adaptation** and a careful balance of speed and quality, innovation, and security.

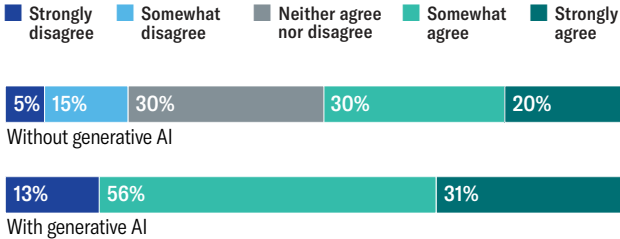
It’s clear that the organizations that will succeed are those that can effectively leverage these **modern software engineering practices** to deliver value quickly, consistently, and securely.

FIGURE 1

Highly Satisfied with Artificial Intelligence

Technology improves developers' lives

I was able to focus on satisfying and meaningful work.



Source: McKinsey & Co. survey, June 2023

by processes that were designed for antiquated technology and less dynamic times.

More than anything, organizations must build an infrastructure and processes that are flexible enough for an era where change will be constant. “The future of applications and software will probably be much different than what we see today,” says Adam Burden, global innovation lead for Accenture plc, a global professional services firm headquartered in Dublin. “I can see a world where data-driven development starts to take over, and instead of long software programs with lots of code, you use things like knowledge graphs. If you don’t invest in the right architecture and implement the proper governance today, you will find yourself falling further and further behind.”

This paper explores the transformation of software engineering, examining how companies can adopt the four key pillars of modern software development to innovate, adapt, and succeed in a landscape where speed is paramount. The report will probe the change management, people, and technology issues that companies must address as they transition from legacy practices and tools to a fundamentally different method of delivering software.

Boosting Efficiency via Artificial Intelligence

AI and machine learning (ML) are adding to the revolution in every aspect of software development, from code generation to testing and deployment.

AI-powered tools can suggest code completions or even generate entire functions based on natural language descriptions. AI can help generate test cases, predict which areas of code are most likely to contain bugs, and even heal some of these defects. AI can analyze historical project

data to anticipate potential issues or delays in the software development process.

At Boston Consulting Group Inc. (BCG), a management consulting firm based in Boston, AI-powered automation has been extended to the critical task of software testing. “Software isn’t done until it’s tested and any defects are resolved and retested,” says Ryan Bolt, global IT software engineer, director, and Center of Excellence lead at BCG. “You can have the best software application in the world, but if it takes 10 minutes to load, your adoption is going to suffer.”

At the consultancy, programmers have embraced gen AI to write code. Now the organization is exploring test cases for software quality engineers to use gen AI tools to boost their efficiency, as well. “The biggest feedback I get from software testers is they need more time and [quality assurance] capacity, but we don’t have infinite resources,” Bolt says.

Using gen AI helps plug that gap. BCG’s software quality engineers are using tools like ChatGPT and GitHub Copilot to write frameworks for automation code faster and also to test business requirements for different test scenarios. This step has been embedded in the testing process, allowing engineers to get a list of suggested test cases at the click of a button.

“AI can take some of the more tedious parts of our human jobs so we can focus on the more creative solutions that AI can’t do,” Bolt says. “We’re assessing metrics from this approach, including adoption rates and lines of code completed. We’re seeing business value KPIs increase from reduction of time to market, increased velocity in the sprints [a fixed-length time period during which specific work is completed and made ready for review], and reduction in defects.”

AI must be central to such tasks as testing and observability, according to Shawn Landreth, vice president of networking and network reliability engineering for Capital Group Companies Inc., a century-old financial services firm based in Los Angeles.

“A year ago, we had over 13,000 alerts a day. We chopped that down to less than 1,000 alerts a day, but that’s still a lot of alerts for a single team to look at,” he says. “As we evolve our monitoring platforms, we need AI to sit in the middle of it, to add context and fill in gaps so we can determine which issues really call for our immediate attention.”

Embedding Security and Governance

As development speeds increase and systems become more complex, security and governance must be woven into every stage of the development process. Robust data governance measures are especially important when developing applications with AI and ML models.

“As applications get easier to build, folks get a little bit lax,” says Brian Hinshaw, business management principal at telecom giant AT&T, headquartered in Bedminster,

N.J. “You have to come back and check their work. Audits are a big thing internally, and we [also] have third-party audits going on. It never goes away from being a part of the development process.”

Low-code and no-code platforms—tools that allow users to create applications or automate processes without extensive programming knowledge—are democratizing software development, allowing “citizen developers” (i.e., users themselves) to create applications quickly without extensive coding knowledge. “I’ve had some citizen developers build their solutions in a matter of a few days, and now they maintain them,” Hinshaw says. “If you look at a traditional development practice, nothing gets done in a few days except maybe the requirements.”

In this complex environment, companies find themselves challenged to monitor security across the growing array of vulnerabilities or “attack surfaces,” make time for process improvement, and respond to security in a timely manner, according to a June 2024 survey by the SANS Institute, a nonprofit focused on cybersecurity based in Bethesda, Md. **FIGURE 2**

Such challenges mandate that companies integrate security practices early in the development process rather than treat the integration as an afterthought. They must use tools to automatically check for and enforce compliance with relevant regulations and standards.

“Most modern software engineering has started to emphasize a lot more reuse of code, a lot more open-source code, and other internal source libraries,” says Adam Airhart, manager of cybersecurity at United States Steel Corp., a Pittsburgh-based steelmaker. “So if you have security flaws in one area, those flaws are replicated in a lot of places.”

When Airhart was appointed to his current position in 2019, he inherited an environment of antiquated software systems, some over a decade old. The company’s risk profile was unclear, with numerous websites vulnerable to unknown threats.

Airhart began with a simple yet powerful step—introducing version control through an AI-powered development platform. This seemingly small change laid the foundation for a more robust development process. His team then introduced build-and-release pipelines, which are processes that automate the steps involved in software development, eliminating many manual tasks that previously caused headaches for developers.

In addition, U.S. Steel has embraced the fourth pillar, embedding security in development practices. “We have gotten a good bit [outside] our comfort zones with AI moving at the speed of light, and we need to learn as fast as the new AI technology is coming out,” Airhart says. The company established “approval gates,” which are checkpoints in the build-and-release process that require explicit approval

FIGURE 2

Security Challenges

Growing attack surfaces and insufficient time are top issues

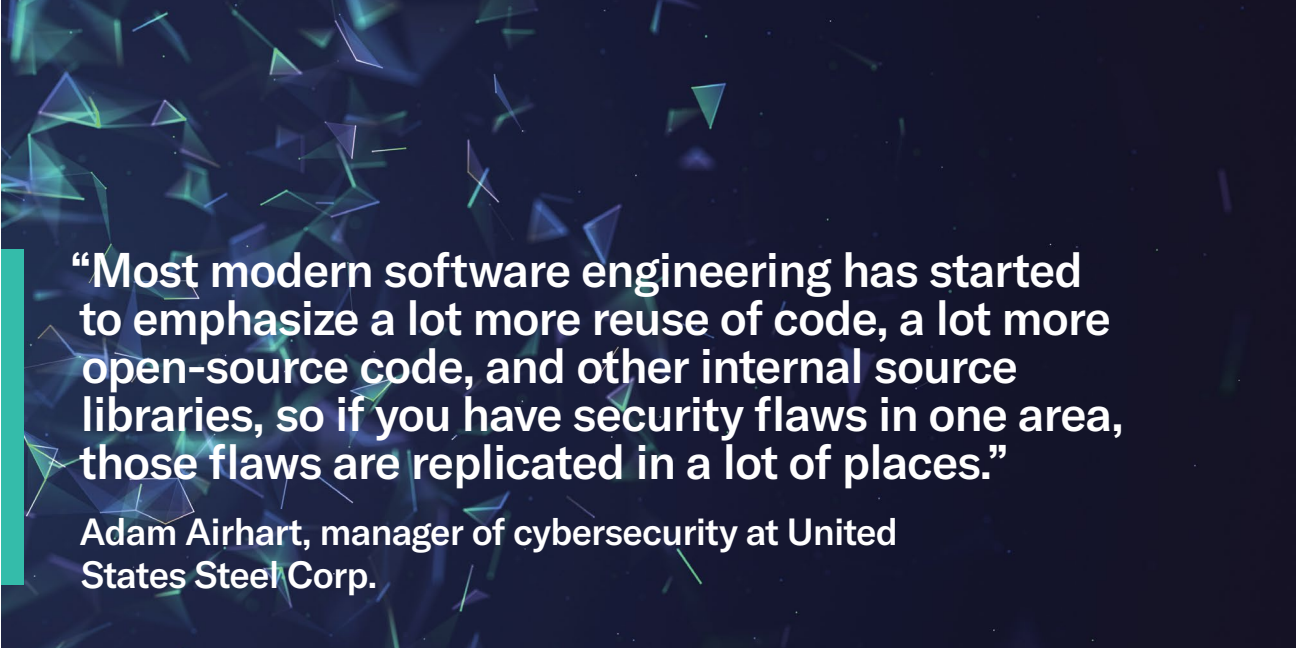
Which of the following would you say are your organization’s most significant security operations challenges? *Select up to three, not in any specific order.*



Source: SANS Institute survey, June 2024

before proceeding to the next stage. “The developers get instant feedback, so they can correct any issues immediately,” Airhart says. While most approval gates at interim steps are automated, approval from a human being is necessary when a developer wants to move from one key phase to another key phase in the process.

As Accenture’s Burden notes, “There’s no substitute for having a really strong set of standards and governance that guide how you do things. The most successful companies have a strong central governance body and some astute architects



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who are helping kind of steer their legions of programmers in the right direction. It’s tricky to get this right—be too strict and you stifle innovation, be too loose and you’re building a lot of future legacy debt. Find a middle ground that is right for your enterprise.”

Improved Visibility

As systems become more complex and distributed, the ability to monitor and troubleshoot effectively becomes crucial. Modern software engineering demands companies move beyond merely collecting data and generating metrics about system health. Companies need observability, a modern investigative approach into the root cause of issues that examines the data and component interactions of distributed systems.

“A decade ago, major components of systems were often built as monoliths that weren’t more complicated than a client layer, a server layer, and a web layer,” Burden says. “Now, with things like microservices, it’s incredibly hard to diagnose when and where something’s going wrong, and even harder to make sure something doesn’t go wrong in the first place. If you want to apply these modern software development practices in these highly distributed systems, you need observability to ensure it runs successfully in production.”

When Capital Group’s Landreth joined the company in 2018, his charge was to introduce a modern approach to observability that could anticipate issues rather than simply respond to them.

“We had a distress protocol, where nothing was done until someone complained about something,” he says. “We were good at putting out fires, but terrible at preventing the fires from starting. Our reputation was questioned because it looked as if we didn’t know our own infrastructure. If your customers have to tell you there’s a problem, you’re already behind the problem by hours, if not longer.”

Landreth says Capital Group needed to step back and not be myopically focused on one portion of the network environment and instead examine all the contributing factors that affected downstream applications and services. “To improve our overall response time, we had to figure out where in the process to layer in automation and where to layer in human intelligence,” he says.

Capital Group implemented a new monitoring platform and processes. Key aspects of improved visibility included using predictive analytics to analyze historical data and identify patterns to predict potential issues before they occur, determine the underlying causes of issues, and reduce the noise from less-critical issues.

In addition, the company hired network reliability engineers who focused on automation and operations. The organization implemented regular operational calls, where functions as diverse as desktop support and the help desk team could explain issues they were seeing. By connecting the dots, Landreth’s team could anticipate issues rather than simply react to them.

“Everything about our approach had been siloed,” he says. “We were not very good at talking with each other. That’s

not acceptable in an IT world where everything touches everything else. We are no longer living in the waffle world of compartmentalization, where you have your square and I have mine. It's a giant bowl of spaghetti. Now when we have an issue, a variety of teams leap into action together to head it off."

Meeting the New Demands

Chief information officers (CIOs) said it was "very important" to build out their operations teams with skills in areas that are paramount to modern software development over the next 12 months, according to a survey by HFS Research, a consultancy in Cambridge, United Kingdom. These skills included infrastructure operations (cited by 59%), cloud migration (54%), cybersecurity (52%), and business application development (48%). **FIGURE 3**

The need for such a diverse array of talent reinforces that the traditional software development life cycle, often spanning months or even years, is no longer viable in today's fast-paced business environment.

"If we can't get our new software to market to support a new internet service, then we can't sell the product," AT&T's Hinshaw says. "We've invested a lot of money to get that product to market, and we want the revenue back as fast as we can get it."

During the Covid-19 pandemic, AT&T was able to leverage modern development practices to deploy a low-code solution over a weekend that allowed call center workers to work from home. "We didn't have to spend six months coding this



"You can do DevOps without Agile, but you can't do Agile without DevOps. But if you want to adopt modern software practices, you need both, since they support one another," says Adam Burden, global innovation lead for Accenture plc.

solution," he says. "The lightbulb was coming on." As AT&T embraced more modern development practices, productivity exploded. "We have produced somewhere around 60 to 70 apps a quarter," Hinshaw says. "Before, we might get one or two apps a year."

Promoting speed and agility depends on a combination of technology and processes. The build, test, and deployment processes for applications must be automated to enable frequent, reliable releases. Organizations must make use of technology like microservices architecture, which constructs applications as a collection of small, independent services, enabling faster development and deployment of individual components.

Also central to modern software engineering is Agile development, a methodology for software development that emphasizes flexibility in responding to user needs, and DevOps, a set of practices and cultural philosophies that aim to improve collaboration between software development (Dev) and IT operations (Ops). These complementary methodologies emphasize collaboration, flexibility, and continuous delivery, which allow organizations to respond quickly to changing requirements and market conditions.

"DevOps is an umbrella term for all the tooling that is used for the software development life cycle, for running the systems, and for development and operations," Burden says. "You can do DevOps without Agile, but you can't do Agile without DevOps. But if you want to adopt modern software practices, you need both, since they support one another."

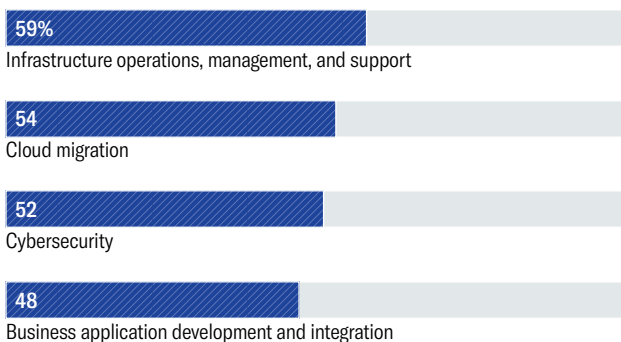
More than three-quarters (83%) of developers were involved in DevOps-related activities for a wide range of software development activities, according to the State of CI/CD Report 2024, published by the Continuous Delivery Foundation, a part of the Linux Foundation in San Francisco. These tasks ranged from monitoring software and/or infrastructure performance

FIGURE 3

Development, Infrastructure Skills Needed

Chief information officers want to build out operations teams

How important are the following technology skills for your team to build out over the next 12 months? (Responses to "Very Important")



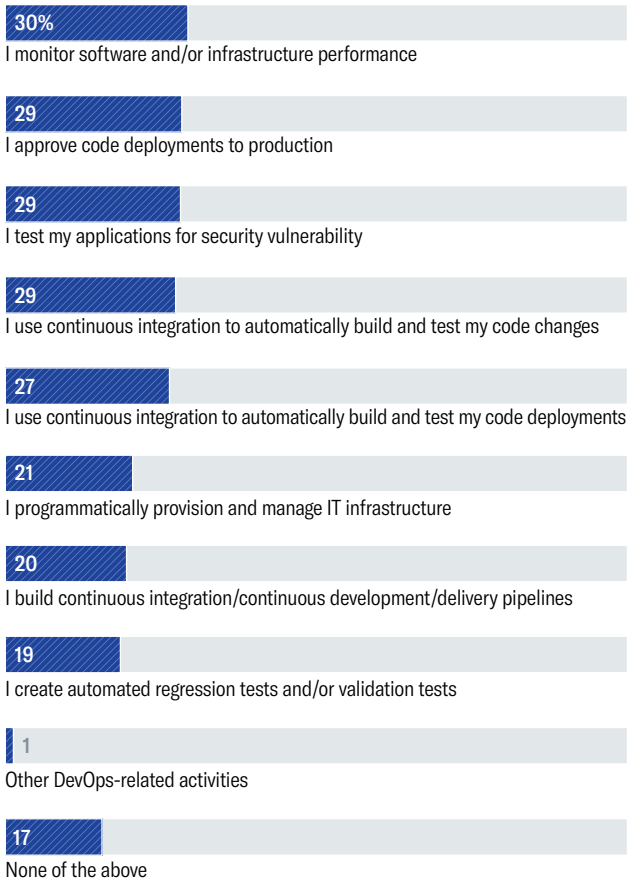
Source: HFS Research survey, June 2024

FIGURE 4

Keeping Tabs

Monitoring software and/or infrastructure is top use of DevOps

Which of the following development activities are you involved in? (Q1 2024)



Source: Continuous Delivery Foundation survey, February 2024

to approving code deployments for production. **FIGURE 4** The survey found that using a broad range of DevOps technologies is correlated with better delivery performance in areas like the time it takes to make code changes or restore services after an outage.

“DevOps is one of the biggest things that makes modern software engineering work,” says Burden. “Those are the tools that help you move an application all the way from checking its source code into a repository through your testing into deployment scripts, and ultimately into production. You can’t introduce new practices like Agile without having an infrastructure and architecture that allows you to speed the process up. It’s like a clock without the mechanics behind it.”

Strategies for Adoption

To successfully adopt modern software engineering practices, experts and practitioners advise focusing on several areas.

Get buy-in from management. “The biggest obstacle is changing the culture and behavior,” Forrester’s Lo Giudice says. “At the team level, the people on the ground get really excited. However, I see lots of misunderstanding and resistance at the management level because the executives don’t understand the investment in change management that needs to happen.” He says executives need to understand the principles and values of Agile development, adding that successful companies often use Agile principles to inform their business strategy.

Court buy-in from IT personnel. Despite excitement at lower levels, resistance to change within the IT department is also a challenge to adopting modern software engineering practices. Overcoming that resistance requires a multifaceted approach including communication strategies, training programs, and gradual implementation.

Competing priorities between business needs and cybersecurity requirements can create tension. At U.S. Steel, Airhart’s team had to become salespeople, showcasing the benefits of new practices to skeptical developers. They demonstrated how updating a website could be reduced from 45 minutes of downtime to just 30 seconds using their new infrastructure.

A key revelation came when Airhart realized the importance of developer buy-in. Rather than imposing tools from the top down, his team involved developers in the selection process. They tested various products, ensuring that the chosen tools met both security and developer usability needs.

AT&T implemented “a bug bounty” that rewarded developers for fixing issues with software. Hinshaw explains, “We put the results on a board that everyone could see, and people started to say, ‘I want some of that attention, too.’”

Balance innovation and security. As organizations embrace AI and automation to improve efficiency, they must also ensure they’re maintaining security and compliance. This task requires a delicate balance between innovation and risk management. At the same time, workers must be assured that new technology, like AI, will augment their jobs rather than replace them.

“We’re creating safe spaces for people to innovate and try new things,” BCG’s Bolt says. “And we’re not just trying to cram metrics down your throat and make you work faster so that we can eliminate half these jobs. We’re not trying to eliminate any positions. We’re trying to do more with the same number of people we now have.”

Check in often with customers. To ensure that modern software engineering practices are delivering value, organizations need to focus on KPIs such as development



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speed and time to market, quality metrics and defect rates, customer satisfaction and user experience, security compliance, and incident rates.

Continuous improvement should be at the heart of any modern software engineering practice. This mission involves regular assessments, feedback loops, and a willingness to adapt to new technologies and methodologies. “We are aggressively customer-focused, so we do a lot of polling of our customers to ensure our solutions meet their requirements,” Bolt says.

Future-proof values and principles. “Being fit for the future doesn’t mean building something today that will last for 20 years,” Lo Giudice says. “It means being efficient and effective in changing. We don’t know what’s coming around the corner.”

He cites the basis of Agile development as an instructive change format. “Agile transformation is a process that never ends,” Lo Giudice says. “You don’t have to be religiously tied to everything in the Agile manifesto. But companies have been successful by letting the values and principles of Agile development influence their business strategy—if you hire a workforce that is made up of problem solvers rather than process masters.”

Conclusion

Modern software development is undergoing profound change as organizations adopt new practices to provide the speed, agility, and innovation that is required by the changing business world.

“The ability to change should be a high priority for all managers and executives,” Lo Giudice says. “But to be very fast at changing, tweaking, and adapting the business to the changes in external business conditions, Agile alone won’t be enough. You’re going to have to be very effective in putting in place reliable engineered platforms so that digital teams are enabled to build, manage, and deliver the software you need, at the time the business needs it.”

The transformation requires a shift in organizational commitment to continuous learning and adaptation and a careful balance of speed and quality, innovation, and security.

“Without modern software practices, you simply can’t keep up with the velocity and complexity of change while maintaining the quality you need,” Burden says. “If you cut to the core, this is about providing a better user experience by producing business software that functions at a similar level that people have come to expect from consumer software. I don’t think it’s an unreasonable expectation for people to have that their business software is just as feature-rich, reliable, and regularly improved as their favorite streaming service.”

As we look to the future, it’s clear that the organizations that will succeed are those that can effectively leverage these modern software engineering practices to deliver value quickly, consistently, and securely.

“No one knew three years ago that gen AI and ChatGPT were going to exponentially change the world,” says Lo Giudice. “And no one knows what the next change will be. But your software development practices need to be ready for anything, because change is a certainty.”



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